



1.1 Number systems, rounding and estimation

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Numbers are classified as natural, integer, rational (terminating/repeating decimals, expressible as p/q) or irrational (non-terminating, non-repeating, e.g. $\sqrt{2}$, π). Real numbers can be written in standard form $a \times 10^k$ ($1 \leq a < 10$). Approximation involves rounding to a given number of decimal places or significant figures, and accuracy is measured using percentage error = $|\text{approximate value} - \text{exact value}| / |\text{exact value}| \times 100\%$.

Worked example: follow the reasoning

A construction certificate states a plot's area as $8,362,450 \text{ m}^2$. (a) Write this in standard form correct to 3 significant figures. (b) Calculate the percentage error of this approximation.
Solution: (a) $8.36 \times 10^6 \text{ m}^2$. (b) Approx value = $8,360,000$. Error = $|8,360,000 - 8,362,450| = 2450$. Percentage error = $2450 / 8,362,450 \times 100\% \approx 0.0293\%$.

Quick check · multiple choice

Which of the following is an irrational number?

- A 0.25
- B $\sqrt{16}$
- C π
- D $7/2$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A construction survey originally records a plot's area as $8,362,450 \text{ m}^2$. (a) Write this measurement in standard form correct to 3 significant figures. (b) Calculate the percentage error of this approximation, giving your answer to 3 significant figures. (c) State one reason an engineer might choose to round a measurement before using it in further calculations. [4]

Final answer: _____



Answers and explanation

Question 1 [4]

(a) $8.36 \times 10^6 \text{ m}^2$. (b) Approx = 8,360,000; error = 2450; percentage error = $2450/8,362,450 \times 100 \approx 0.0293\%$. (c) Rounded values are easier to communicate/report and are sufficiently accurate for practical engineering purposes, avoiding false precision.

Quick check answer

C. π is non-terminating and non-recurring, so it cannot be written as a ratio of integers.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.